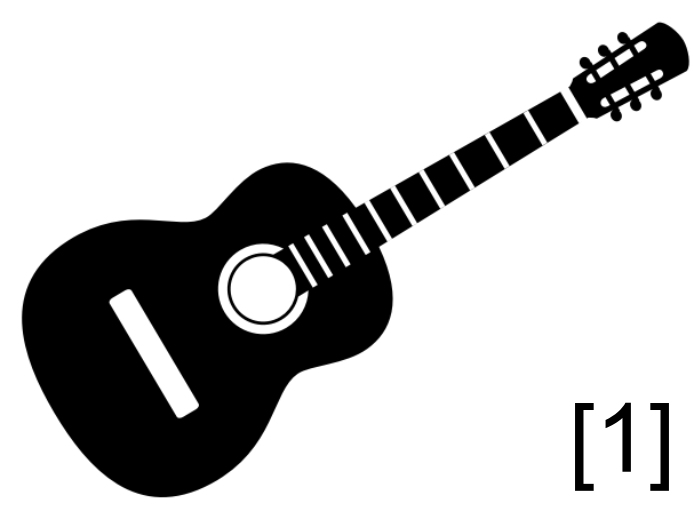




# Strike a Chord

Merlin Gogolin

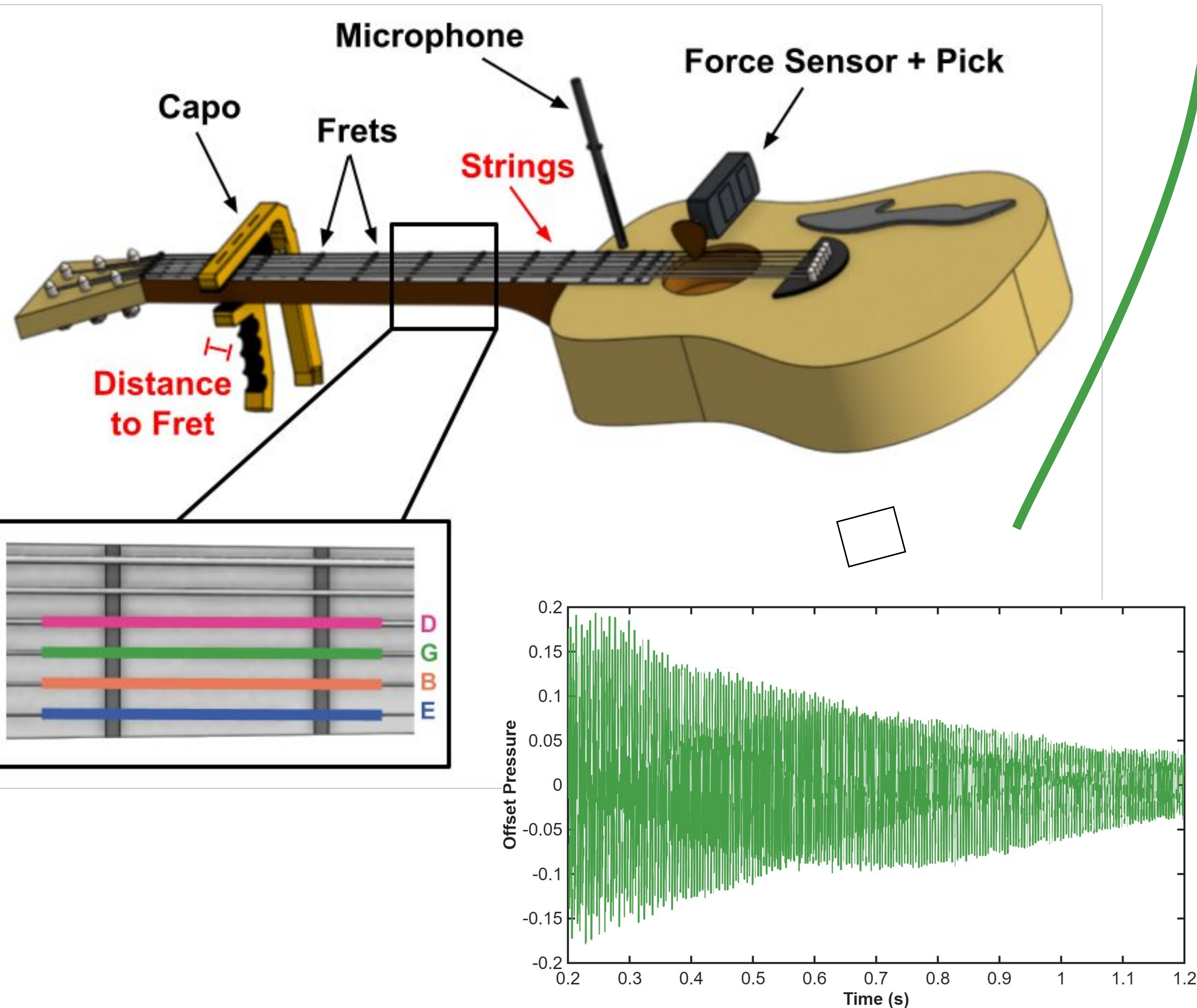
2.671 Measurement and Instrumentation



## Abstract

Guitarists are often told to press down the strings as close to the fret as possible to produce the best sounding chords, which can be difficult to accomplish consistently when fingering complex chords. This study analyzed the relationship between finger distance to fret and sound quality across 4 strings (high **E**, **B**, **G**, and **D**) of a standard acoustic guitar. A capo was used to apply a constant force across 5 discrete distances to the second fret while the respective strings were plucked with a controlled force. Sound was recorded with a microphone and an FFT was performed to calculate the Pitch and Spectral Flatness. A consistent increase in Spectral Flatness occurred as distance from the fret increased. Pitch remained constant across the 3 closest distances but became flatter past a certain threshold. The results confirm that guitarists should indeed place their finger as close to the fret as possible.

## Experimental Setup



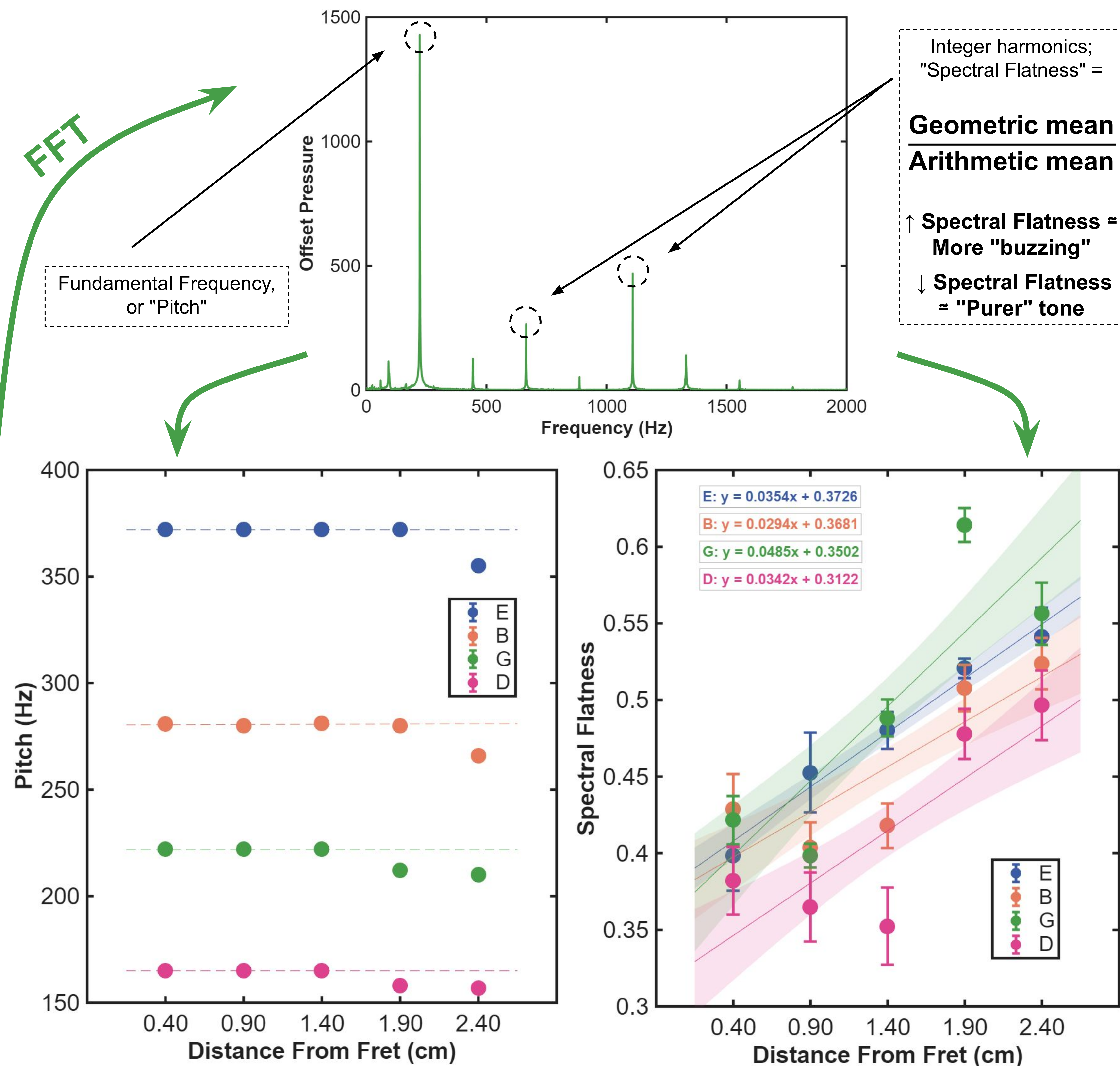
## Acknowledgements

Thank you to Dr. Lachner and Prof. DiGenova for their support and advice throughout this project.

## References

[1] <https://freemvg.org/guitar-black-and-white-illustration>

## Sound Quality Across Distances



## Conclusions

- In general for the 4 highest strings of an acoustic guitar, Spectral Flatness increases as finger distance to the second fret increases
- For the (high) **E** and **B** strings, Pitch starts to get flatter after 1.90 cm from the fret
- For the **G** and **D** strings, Pitch starts to get flatter after 1.40 cm from the fret
- All 4 strings produce a similarly low Spectral Flatness at 0.40 cm from the fret, confirming that fingering close to the fret produces a consistently "pure" tone
- The Spectral Flatness of the **G** string is the most sensitive to finger distance, while the Spectral Flatness of the **B** string is the least sensitive to finger distance